EARTH SYSTEM COMPONENTS
12-GEO-M-AG02

Lectures: Miguel Mahecha

Exercises (2026): David Montero & Miguel Mahecha

Institute for Earth System Science and Remote Sensing



UNIVERSITÄT
LEIPZIG

Before Christmas: atmosphere, hydrology, carbon

Block	L	Date	Topic
Intro	L01	13.10.2026	Course overview and organisation
I · Atmosphere	L02	20.10.2026	Atmosphere I: concepts and key variables
	L03	27.10.2026	Atmosphere II: radiation balance
	L04	**03.11.2026**	**Atmosphere III: structure of the atmosphere (today)**
	L05	10.11.2026	Atmosphere IV: buoyancy forces
	L06	17.11.2026	Atmosphere V: circulation
II · Terrestrial hydrology	L07	24.11.2026	Terrestrial hydrology I: water-cycle concepts
	L08	01.12.2026	Terrestrial hydrology II
	L09	08.12.2026	Terrestrial hydrology II: measurements, Budyko framework
III · Biogeochemistry	L10	15.12.2026	Biogeochemistry I: element cycles – the carbon cycle

… biogeochemistry, classification, special topics

Block	L	Date	Topic
–	–	*22.12.–02.01.*	*Christmas break*
III · Biogeochemistry	L11	05.01.2027	Biogeochemistry II: terrestrial biogeochemistry
IV · Classification & special topics	L12	12.01.2027	Climate classification systems and their limitations
	L13	19.01.2027	Atmospheric rivers
	L14	26.01.2027	Forest damage and climate change – guest lecture: Prof. Henrik Hartmann
	L15	02.02.2027	Climate classifications and vegetation
–	–	*Feb 2027*	*Oral exams (dates tbc)*

Remarks on the exercises and exam...

- The portfolio will be an “exam entry prerequisite”, i.e. you need to submit a complete collection of all questions and exercises to go into the exam.
- The prerequisite will not be evaluated, the mark is set only in the exam!
- By the end of the lecture period, we expect from you *a single Jupyter Book that collected all exercises including the short presentations you had to do*
- The exam is oral and we will ask questions about any material seen, maybe also asking you to explain an element of your answers or notebooks

RECAP … REMEMBER …

Questions we need to ask:

1. How much radiant energy does the Sun emit?
2. How much of it reaches the Earth?
3. How much total radiation does the Earth absorb?
4. How much does the Earth radiate in return?
5. What is the temperature of the Earth?



picture-alliance/dpa

How much energy does the Sun emit?

- We know the temperature *T* of the Sun and can apply Stefan–Boltzmann law:

$$
\begin{aligned}
E_{\text{sol}} &= \sigma T^4 \\
&= 5,6704 \cdot 10^{-8}\ [\text{W m}^{-2}\text{K}^{-4}] \cdot (5764,5\ [\text{K}])^4 \\
&= 62607,9\ [\text{kW m}^{-2}]
\end{aligned}
$$

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- … and with this the full energy emitted by the surface of the Sun:

$$\begin{aligned} \Phi_{\text{sol}} &= E_{\text{sol}} \cdot 4 \cdot \pi \cdot r^2 \\ &= 62607,9\ [\text{kW m}^{-2}] \cdot 4 \cdot \pi \cdot (6,96 \cdot 10^8\ [\text{m}])^2 \\ &= 3,81 \cdot 10^{26}\ [\text{W}] \end{aligned}$$

Formula sheet I-7, I-8

Questions we need to ask:

1. **How much radiant energy does the Sun emit?**
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picture-alliance/dpa

How much of it reaches the Earth?

- Total energy emitted by the Sun is constant!

$$\Phi_\odot = I_\odot \cdot 4\pi r_\odot^2$$

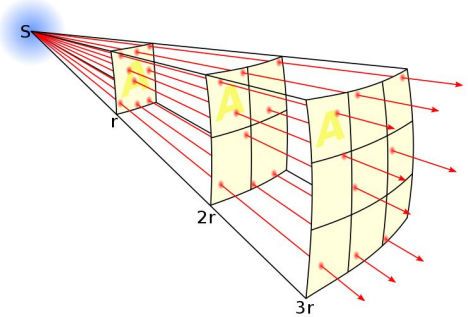
How much of it reaches the Earth?

- Total energy emitted by the Sun is constant!

$$\Phi_\odot = I_\odot \cdot 4\pi r_\odot^2$$

- Yet, with increasing *distance* r_{SE} it is spread over a larger area, so the radiant flux density (power per area) decreases:

$$I_0 = \frac{\Phi_\odot}{4\pi r_{\mathrm{SE}}^2}$$



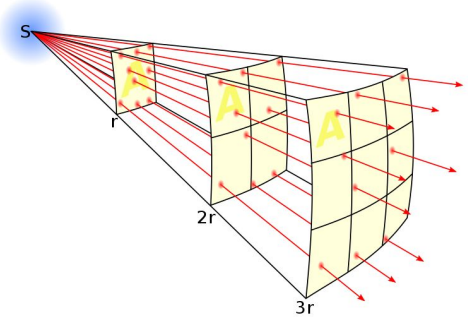
How much of it reaches the Earth?

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- Yet, with increasing *distance* r_{SE} it is spread over a larger area, so the radiant flux density (power per area) decreases:

$$I_0 = \frac{\Phi_\odot}{4\pi r_{\mathrm{SE}}^2}$$



- At the Earth's distance, this flux density is the solar constant:

$$= \frac{3.81 \times 10^{26}\,[\mathrm{W}]}{4\pi \cdot (1.496 \times 10^{11}\,[\mathrm{m}])^2}$$

$$\approx 1355\,[\mathrm{W\,m^{-2}}] \quad (\text{measured: } 1361\,[\mathrm{W\,m^{-2}}])$$

Formula sheet I-9

Question from last lecture: *Do we need the "full sun" calc?*

$$I_0 = \frac{\Phi_\odot}{4\pi r_{\mathrm{SE}}^2} = \frac{I_\odot \cdot \cancel{4\pi} r_\odot^2}{\cancel{4\pi} r_{\mathrm{SE}}^2} = \frac{\sigma T_\odot^4 \cdot \cancel{4\pi} r_\odot^2}{\cancel{4\pi} r_{\mathrm{SE}}^2} = \sigma T_\odot^4 \left(\frac{r_\odot}{r_{\mathrm{SE}}}\right)^2$$

$$= 5.670 \times 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}] \cdot (5772\,[\mathrm{K}])^4 \cdot \left(\frac{6.957 \times 10^8\,[\mathrm{m}]}{1.496 \times 10^{11}\,[\mathrm{m}]}\right)^2 \approx 1361\,[\mathrm{W\,m^{-2}}]$$

$\Phi_\odot = I_\odot \cdot 4\pi r_\odot^2$ total solar power [W] (I-8)

$I_\odot = \sigma T_\odot^4$ flux at solar surface [$\mathrm{W\,m^{-2}}$] (I-7)

Start from the long route (I-8 + I-9), substitute $\Phi_\odot = \sigma T_\odot^4 \cdot 4\pi r_\odot^2$, and watch 4π cancel. The result is dimensionally correct bec $(r_\odot/r_{\mathrm{SE}})^2$ is dimensionless – dividing by r_{SE}^2 alone would give [$\mathrm{W\,m^{-4}}$].

What does this mean for a solar panel on your roof?

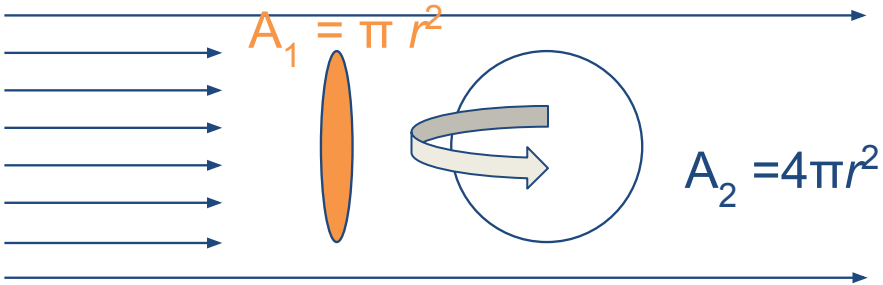
Quantity	W m⁻²	What it means
Solar constant I_0	**1361**	top of the atmosphere, perpendicular incidence
Clear sky at noon, surface	**≈ 1000**	standard test condition for PV ratings (“watt-peak”)
Mean at top of atmosphere, $\bar{I} = I_0/4$	**≈ 340**	averaged over the sphere, day and night
Mean absorbed at the surface (global)	**≈ 161**	after reflection and absorption in the atmosphere
Annual mean in Germany, horizontal	**≈ 124**	1086 kWh m⁻² a⁻¹ ÷ 8760 h (DWD, 1991–2020)

Solar panel efficiency ≈ 20 % → ≈ 25 W m⁻² of electricity, averaged over day, night and seasons
→ ≈ 220 kWh per m² of panel and year
A household using ≈ 3500 kWh a⁻¹ needs ≈ 16 m² of panels (about 8 modules)

Irradiance: DWD, mean 1991–2020; panel values rounded

So what do we effectively get?

- The radiation flux intercepted by the Earth corresponds to the “shadow” with area A_1



- But as the Earth rotates, this energy is spread over the entire surface – four times larger: A2

$$\overline{I} = \frac{I_0}{4} = \frac{1361\,[\mathrm{W\,m^{-2}}]}{4} \approx 340\,[\mathrm{W\,m^{-2}}]$$

Fermi question: how much solar panel area for Germany?

Germany’s gross electricity consumption in 2025: ≈ 526 TWh
Solar energy reaching 1 m² per year: ≈ 1100 kWh
Panel efficiency: ≈ 20 %

→ What area of solar panels would cover the entire annual demand?
→ Compare with the area of Germany (≈ 357 600 km²) – and with the area of Leipzig (≈ 300 km²).

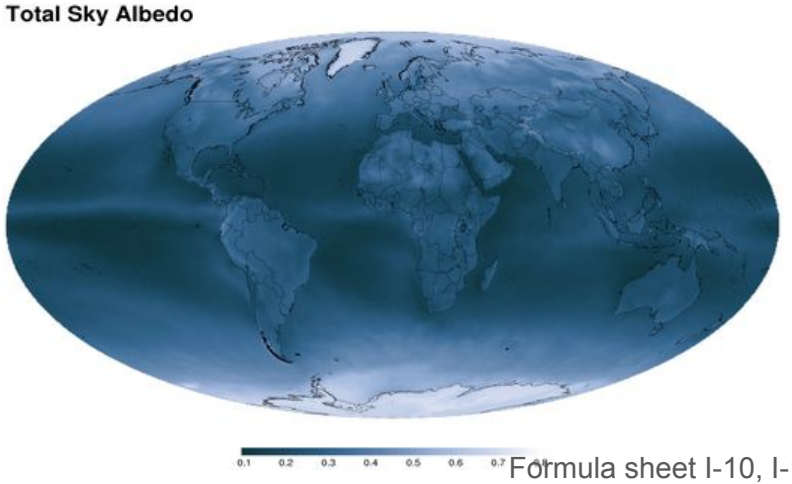
Consumption: Umweltbundesamt (2026), Bruttostromverbrauch 2025. Irradiance: DWD.

So what do we effectively get?

- But the Earth reflects part of the radiation (albedo, approx. 30 %)
- So, on average, the Earth absorbs:

$$\overline{I}_{\text{abs}} = (1-\alpha)\,\frac{I_0}{4} = \frac{(1-0.30)\cdot 1361\,[\text{W m}^{-2}]}{4} \approx 238\,[\text{W m}^{-2}]$$

$\overline{I}$	mean top-of-atmosphere insolation per m^2 of Earth's surface $[\text{W m}^{-2}]$
$\overline{I}_{\text{abs}}$	mean absorbed solar radiation $[\text{W m}^{-2}]$
I_0	solar constant, $1361\,[\text{W m}^{-2}]$
α	planetary albedo, ≈ 0.30 $[-]$
$1/4$	ratio of shadow area πr_{E}^2 to full surface $4\pi r_{\text{E}}^2$



Formula sheet I-10, I-12

Questions we need to ask:

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picture-alliance/dpa

Recall, where we ended: input equals output …

Absorbed incoming radiation = Emission of the Earth

$$I_0\,\pi r_{\mathrm{E}}^2\,(1-\alpha) = 4\pi r_{\mathrm{E}}^2\,\sigma\,T_{\mathrm{e}}^4$$

$$\frac{I_0\,\pi r_{\mathrm{E}}^2\,(1-\alpha)}{4\pi r_{\mathrm{E}}^2} = \sigma\,T_{\mathrm{e}}^4$$

$$\frac{I_0}{4}(1-\alpha) = \sigma\,T_{\mathrm{e}}^4$$

$$\left(\frac{I_0\,(1-\alpha)}{4\,\sigma}\right)^{1/4} = T_{\mathrm{e}} \approx 255\,[\mathrm{K}] \equiv -18\,^{\circ}\mathrm{C}$$



$I_0\,\pi r_{\mathrm{E}}^2\,(1-\alpha)$	absorbed solar power [W]
$4\pi r_{\mathrm{E}}^2\,\sigma\,T_{\mathrm{e}}^4$	emitted longwave power [W]
$\pi r_{\mathrm{E}}^2 / 4\pi r_{\mathrm{E}}^2$	ratio cancels; Earth's size is irrelevant
T_{e}	radiative equilibrium temperature (no atmosphere) [K]

πr_{E}^2 cancels exactly – only the ratio of intercept area to total surface area (= 1/4) remains. This defines $T_{\mathrm{e}} \approx -18\,^{\circ}\mathrm{C}$, the baseline for the greenhouse calculation that follows.

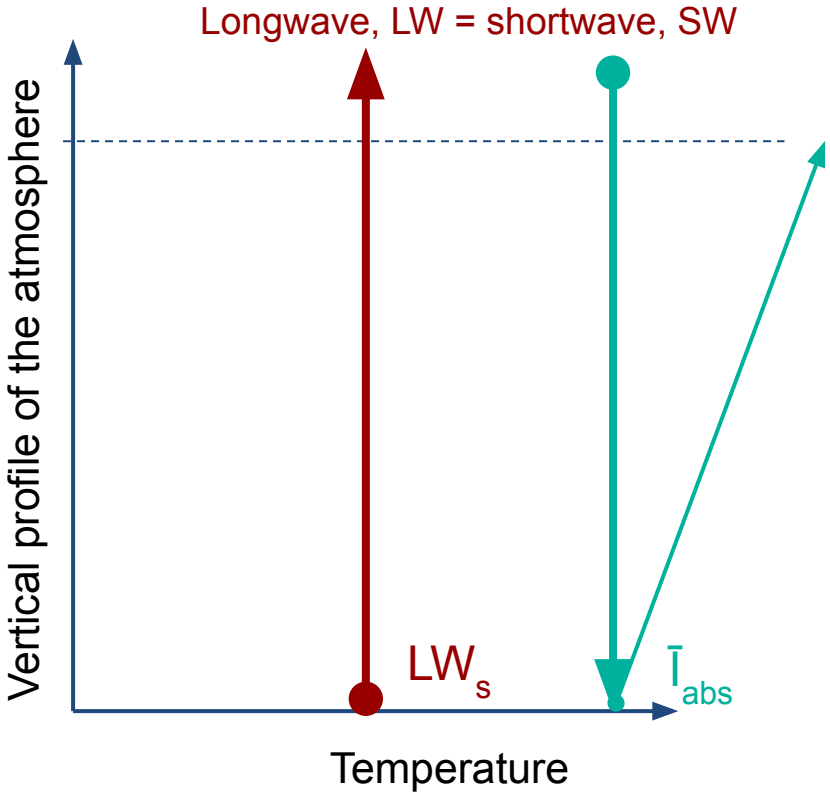
So we concluded: T = -18 °C? Why the hell do we underestimate the temperature so much?

A very simple model of the atmosphere falls short

Absorbed solar irradiation = Longwave emission by Earth

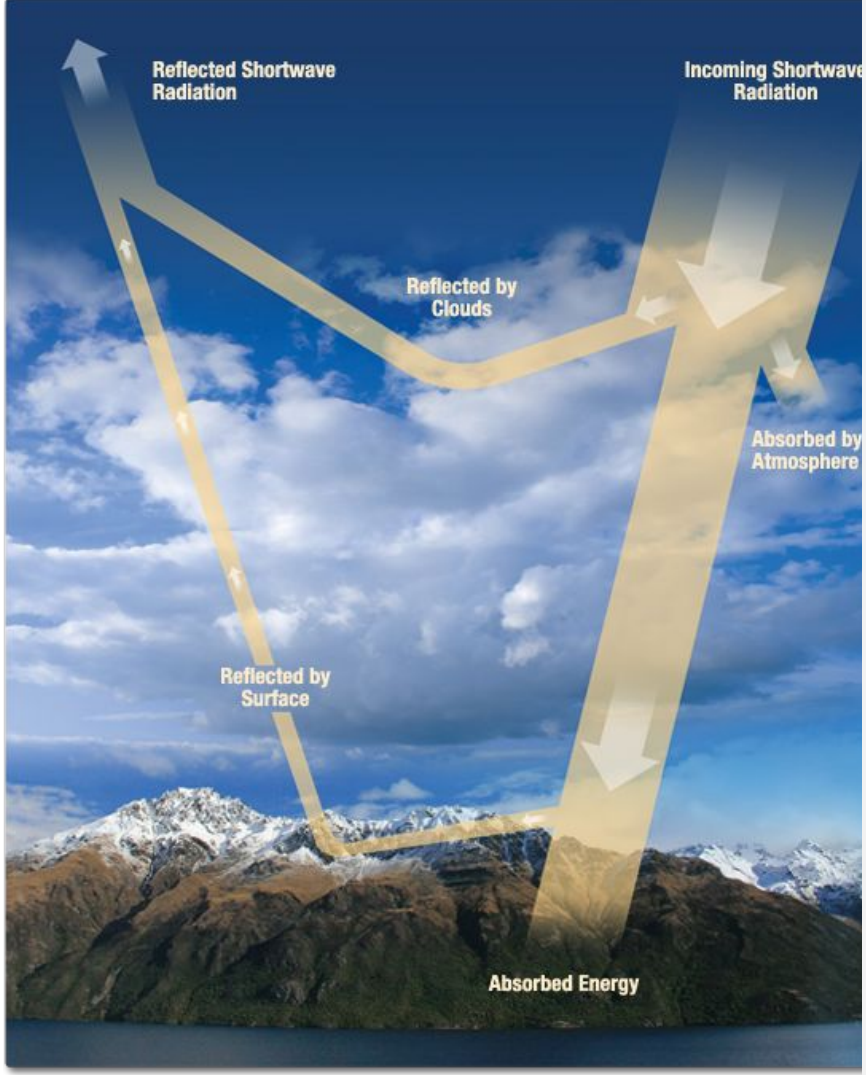
$$\frac{I_0}{4}(1-\alpha) = \sigma T_e^4$$

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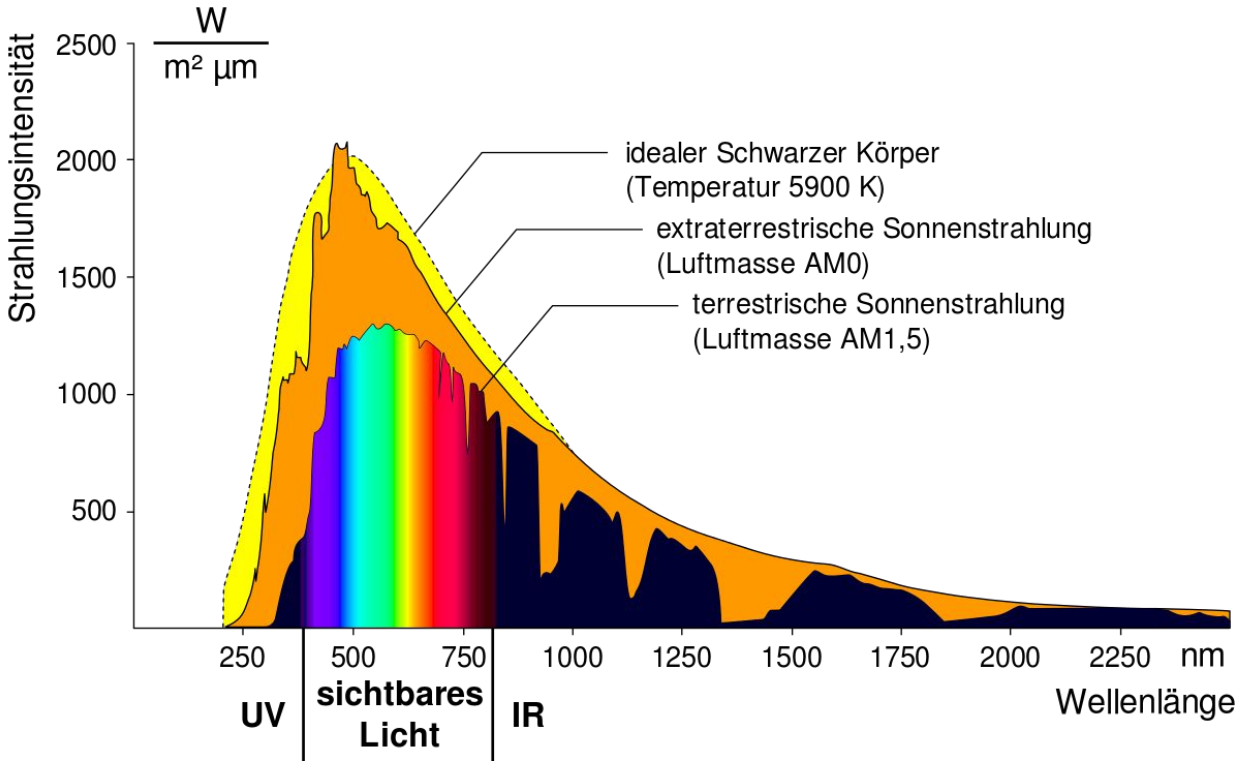
Path of radiation

- This model is only valid if the atmosphere does not absorb any radiation



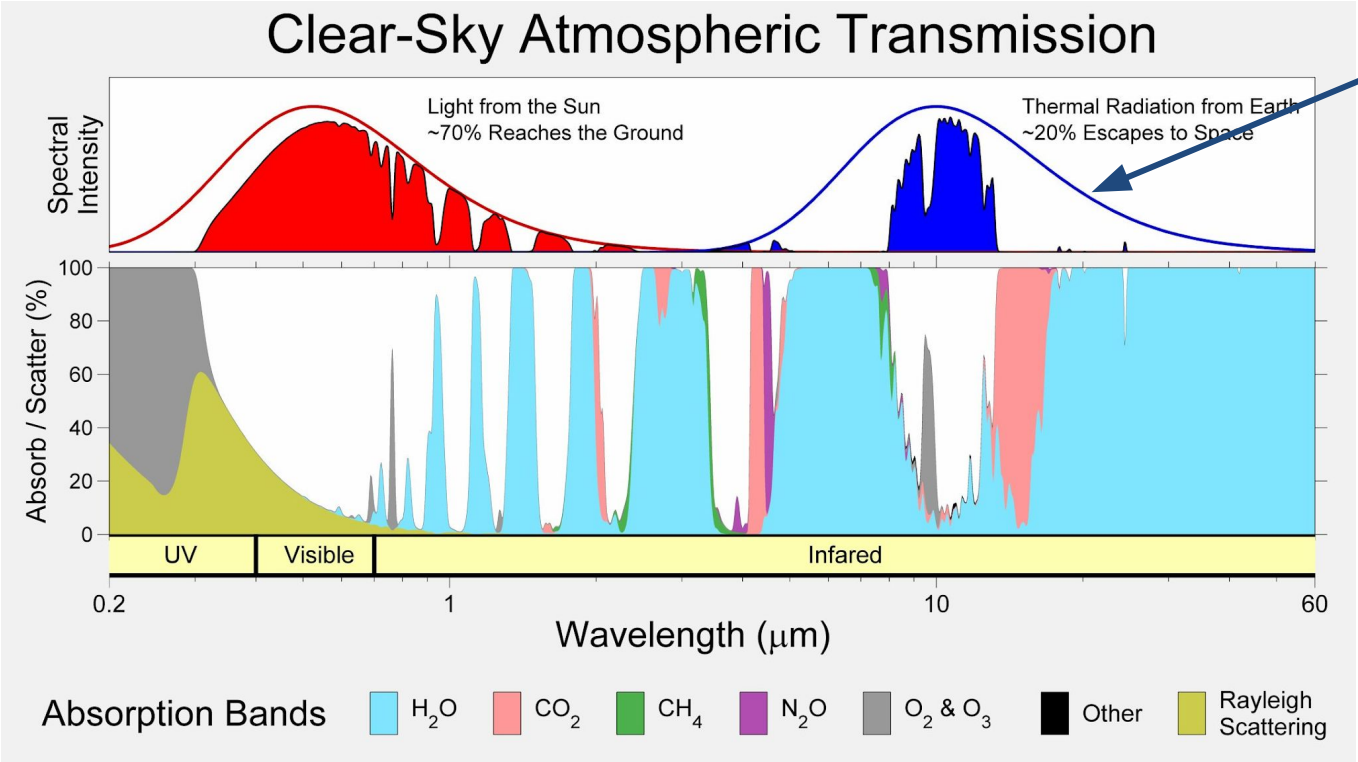
https://science.nasa.gov/ems/13_radiationbudget

Solar radiation above and below the atmosphere



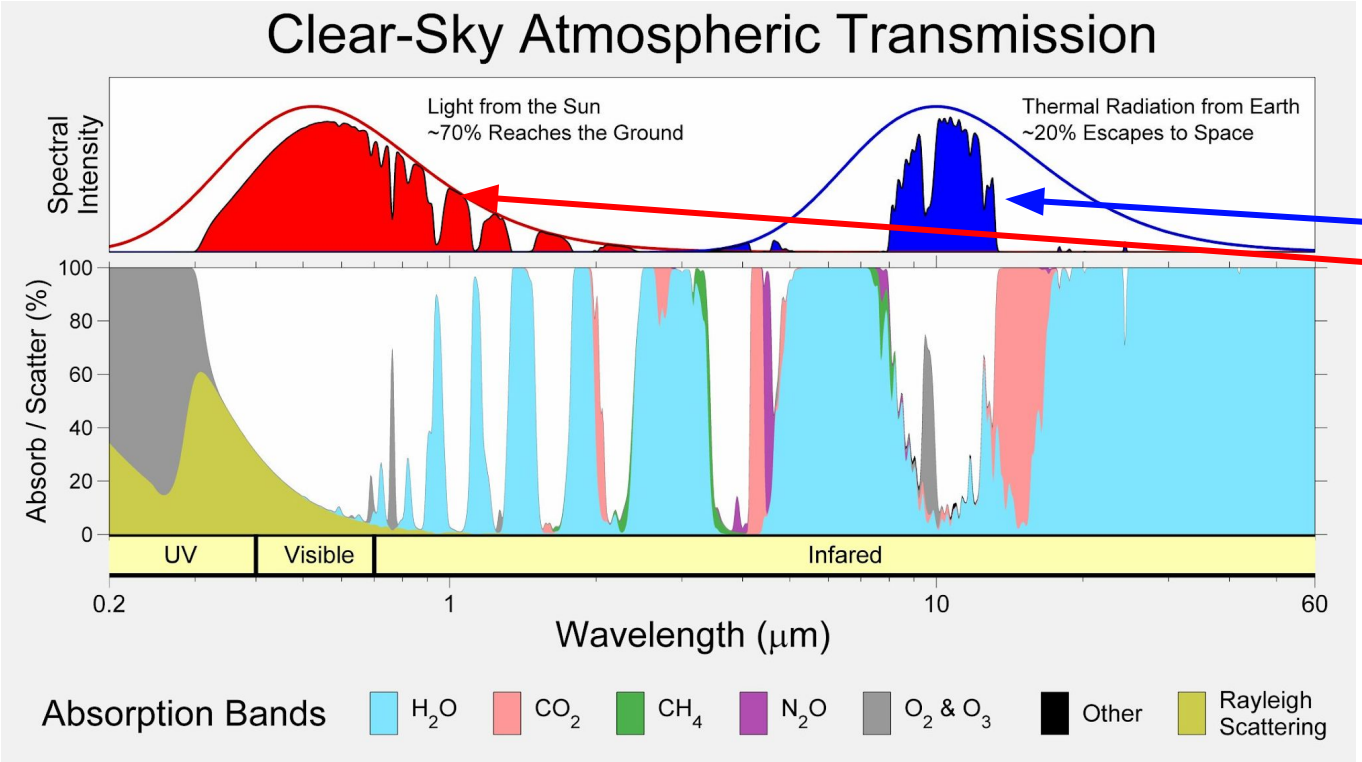
https://de.wikipedia.org/w/index.php?title=Datei:Sonne_Strahlungsintensitaet.svg&lang=de

Transmissivity of the atmosphere



Blackbody emissio
spectra of Sun an
Earth

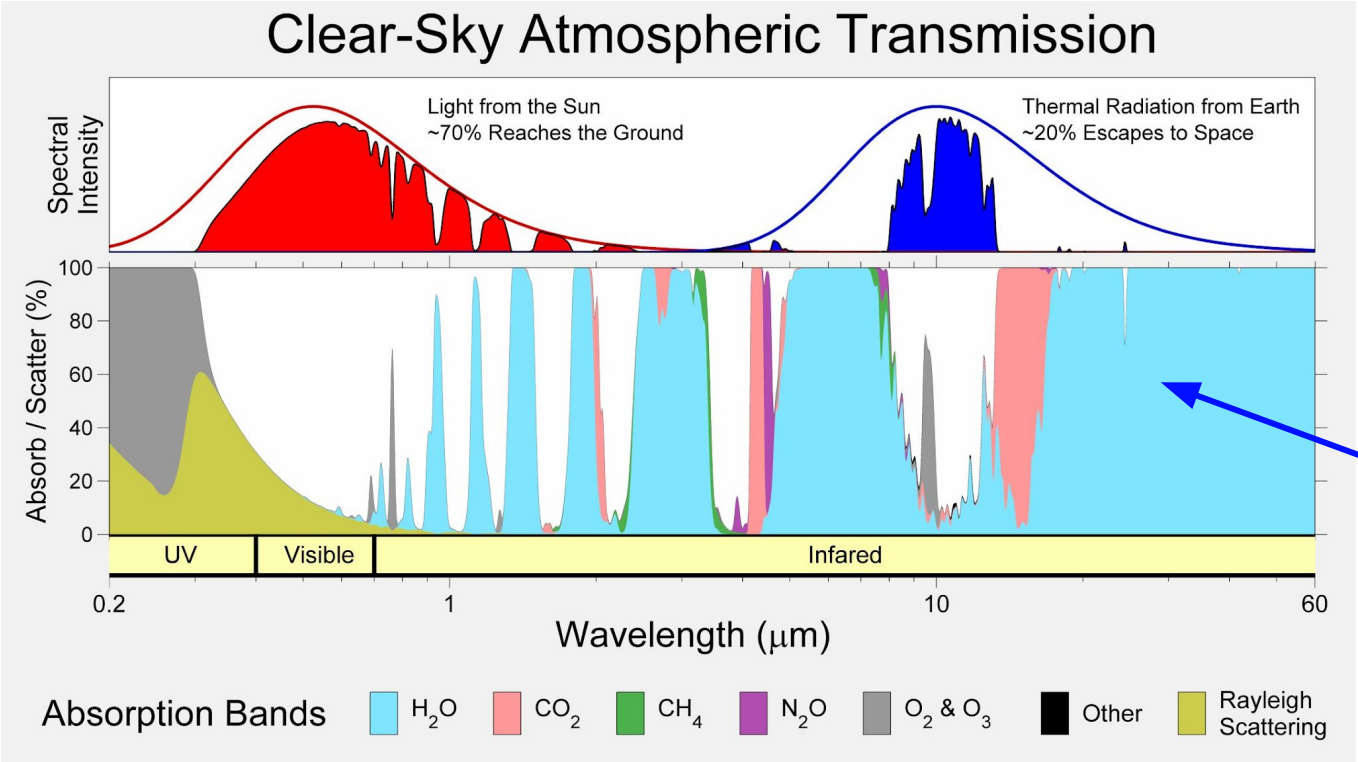
Transmissivity of the atmosphere



Blackbody emissi
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Shaded areas: ac
incoming and
outgoing radiatior

Transmissivity of the atmosphere



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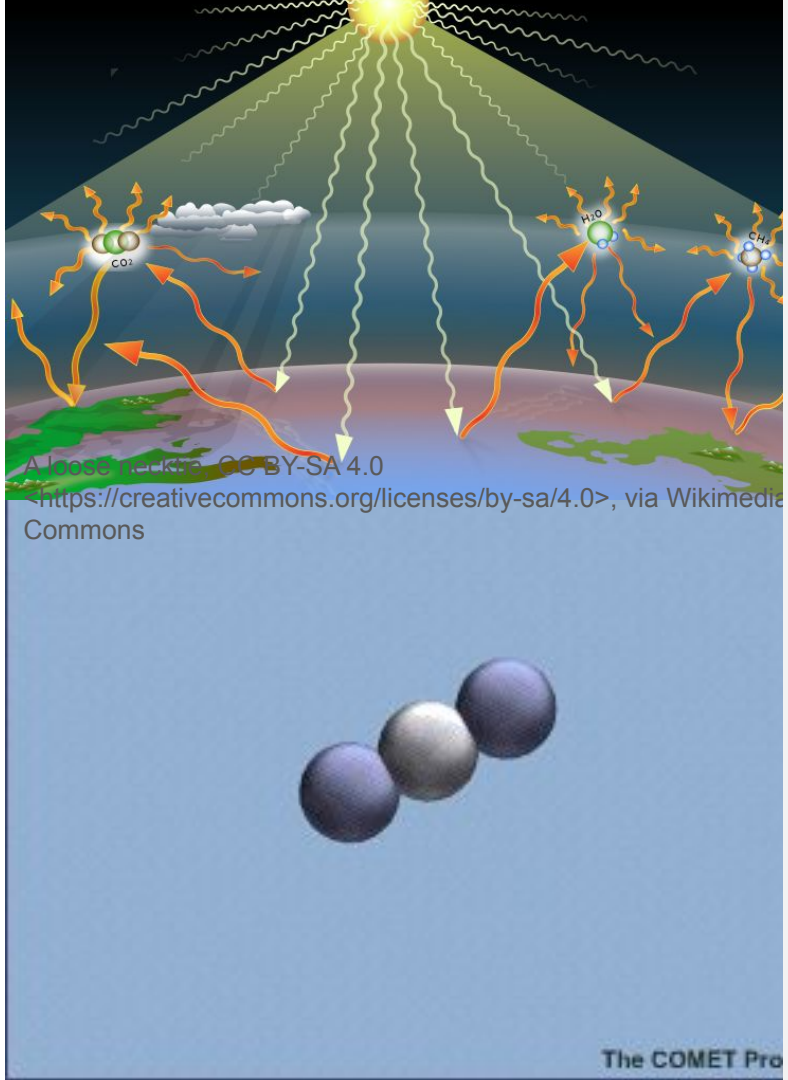
Shaded areas: ac
incoming and
outgoing radiatior

This is because
certain gases
absorb radiation a
specific
wavelengths

Greenhouse gases

- A greenhouse gas absorbs and emits radiant energy in the thermal infrared range.
- Each greenhouse gas has a characteristic absorption spectrum
- Primary greenhouse gases in the Earth's atmosphere are water vapour (H_2O), carbon dioxide (CO_2), methane (CH_4), nitrous oxide (N_2O), and ozone (O_3).
- Average temperature of the Earth without greenhouse gases: approx. −18 °C

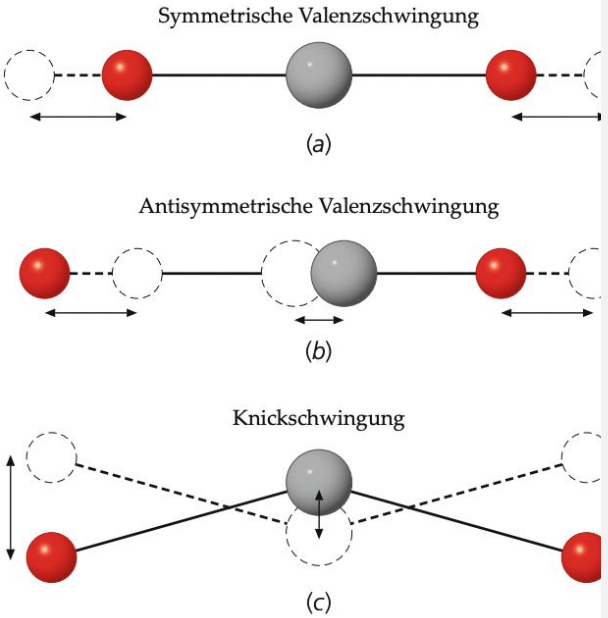
https://scied.ucar.edu/carbon-dioxide-absorbs-and-re-emits-infrared-radiation



A loose nacktie, CC BY-SA 4.0 <https://creativecommons.org/licenses/by-sa/4.0>, via Wikimedia Commons

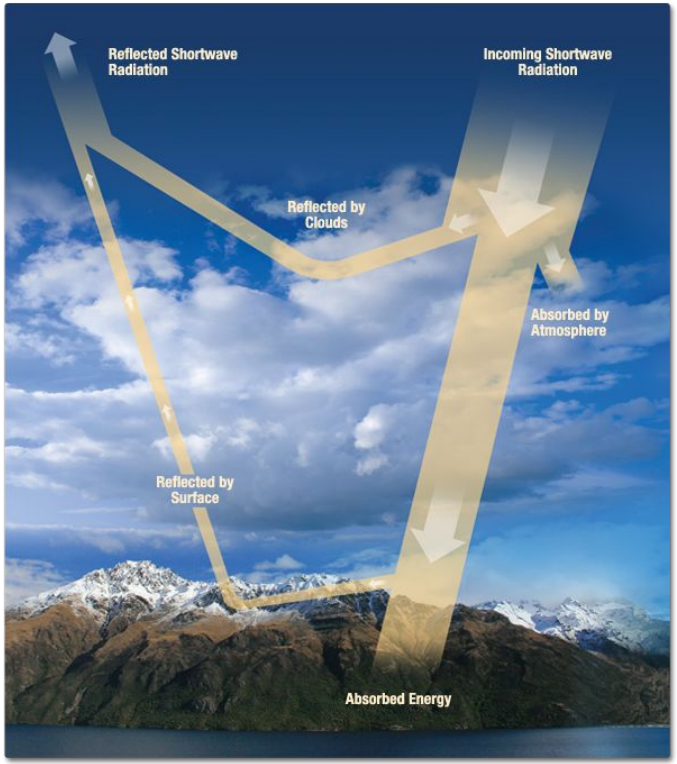
More properties …

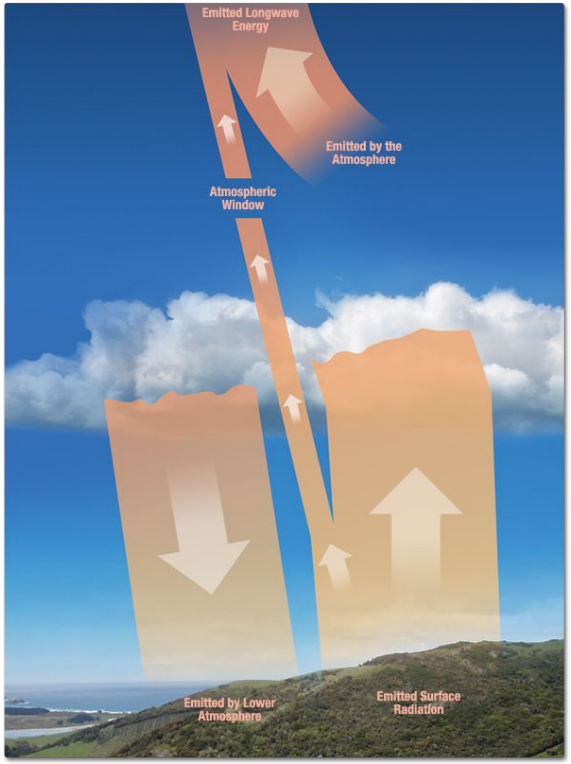
- Greenhouse gases have a variable dipole moment.
- Molecular vibrations change the dipole moment.
- “Static” molecules (e.g. diatomic gases such as O_2 or N_2) do not change their dipole moment through molecular vibrations → transparent to infrared radiation.
- Large molecules (e.g. CFCs, chlorofluorocarbons) → many vibrational modes → many times the global warming potential of CO_2, for example.



Tipler, P. A., Mosca, G., & Wagner, J. (Eds.). Physi
Wissenschaftler und Ingenieure. Springer Spektru

Path of radiation





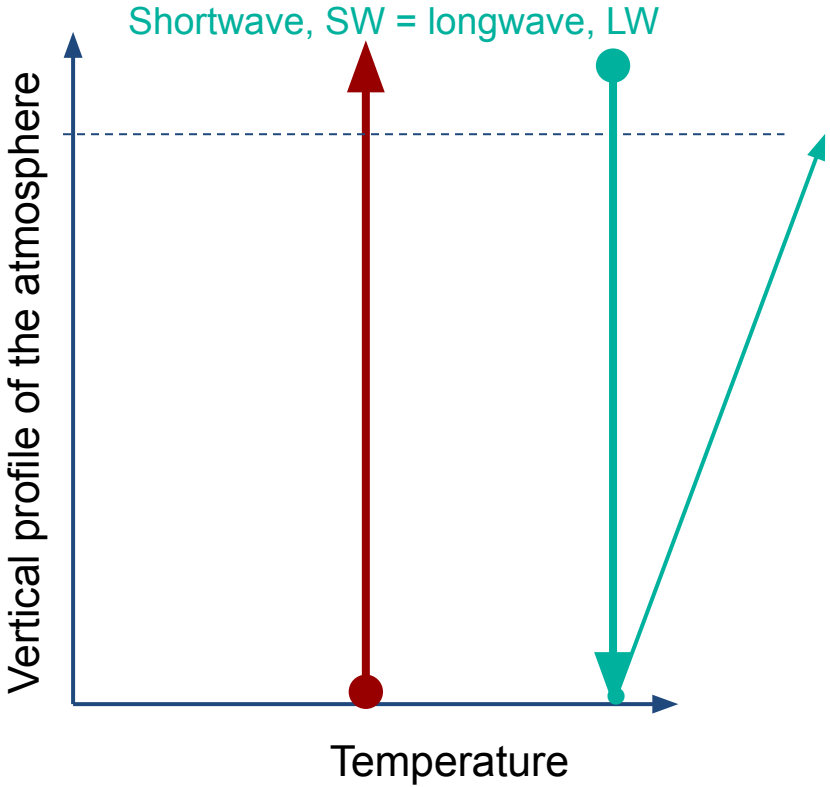
https://science.nasa.gov/ems/13_radiationbudget

Starting point: shortwave in = longwave out

Absorbed solar radiation = longwave emission

$$\frac{I_0}{4}(1-\alpha) = \sigma T_e^4$$

$$\left(\frac{I_0\,(1-\alpha)}{4\,\sigma}\right)^{1/4} = T_e \approx 255\,[\mathrm{K}] \equiv -18\,^{\circ}\mathrm{C}$$

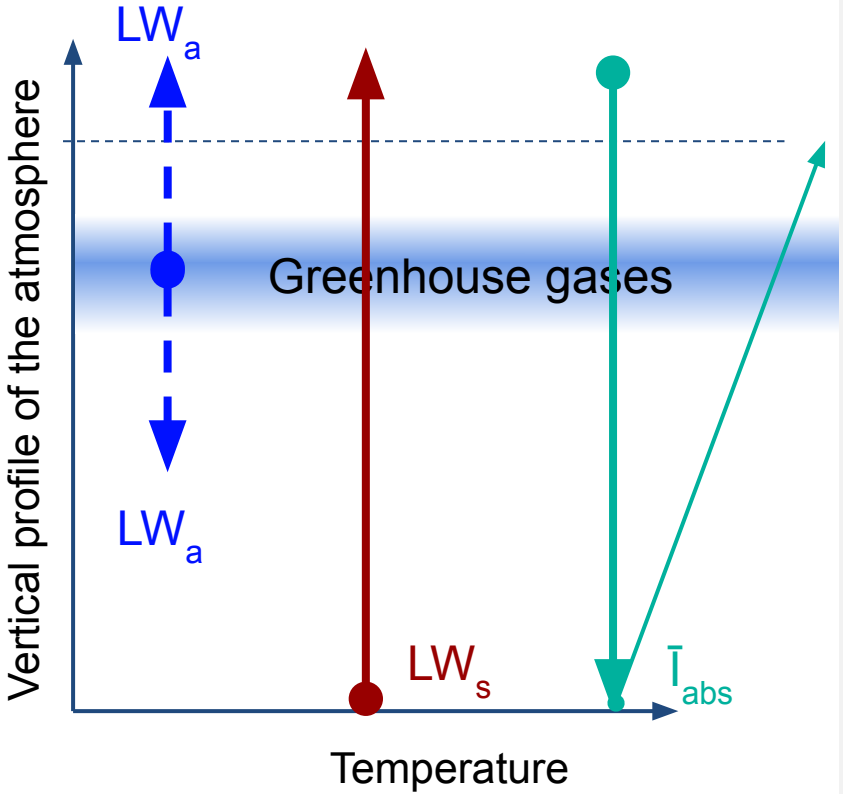


Adding one greenhouse layer

- Now we add greenhouse gases
- Transparent to shortwave radiation
- Absorb all longwave radiation
- Re-emit it upwards and downwards

Atmosphere: $\sigma T_s^4 = 2\,\sigma T_a^4 \quad \Rightarrow \quad T_a^4 = \frac{1}{2} T_s^4$

Surface: $\overline{I}_{abs} + \sigma T_a^4 = \sigma T_s^4$



$\overline{I}_{abs}$	absorbed solar radiation per m^2, $(1-\alpha)\,I_0/4 \approx 238\,[\mathrm{W\,m^{-2}}]$
σT_s^4	longwave emission of the surface $[\mathrm{W\,m^{-2}}]$
σT_a^4	emission of the atmosphere, both upwards and downwards (back radiation) $[\mathrm{W\,m^{-2}}]$
T_s, T_a	temperature of the surface and of the atmospheric layer [K]

Formula sheet I-13

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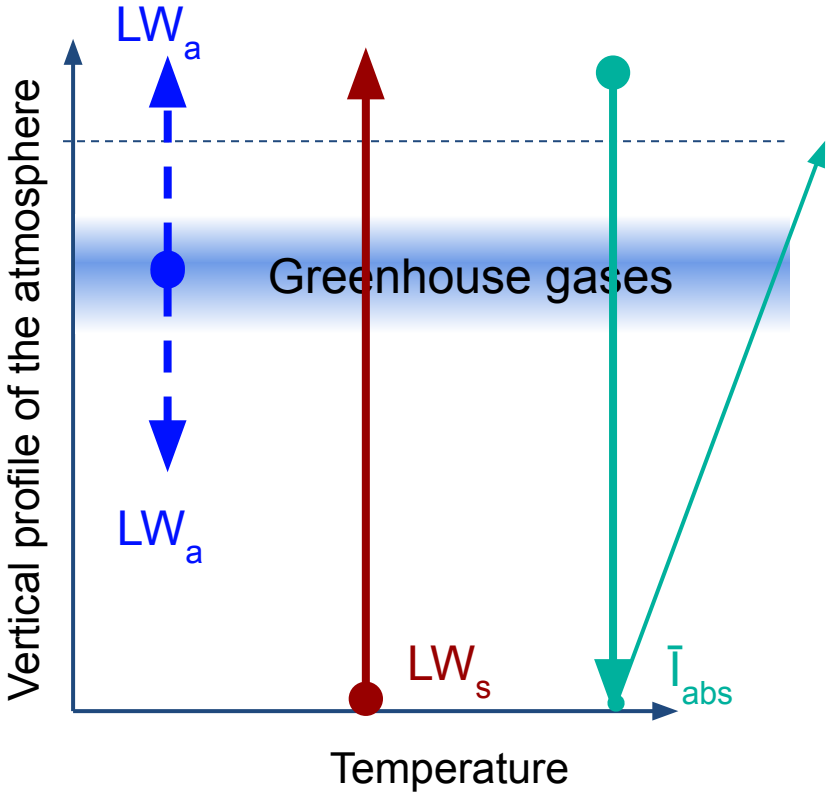
$$\overline{I}_{\mathrm{abs}} + \tfrac{1}{2}\sigma T_{\mathrm{s}}^4 = \sigma T_{\mathrm{s}}^4$$

$$\overline{I}_{\mathrm{abs}} = \tfrac{1}{2}\sigma T_{\mathrm{s}}^4$$

$$T_{\mathrm{s}}^4 = \frac{2\,\overline{I}_{\mathrm{abs}}}{\sigma}$$

$$T_{\mathrm{s}} = \left(\frac{2\,\overline{I}_{\mathrm{abs}}}{\sigma}\right)^{1/4} = 2^{1/4}\left(\frac{\overline{I}_{\mathrm{abs}}}{\sigma}\right)^{1/4}$$

$$= 2^{1/4} T_{\mathrm{e}}$$



$(\overline{I}_{\mathrm{abs}}/\sigma)^{1/4}$ *is exactly* T_{e} *from L03 (formula sheet I-11).*

Formula sheet I-13

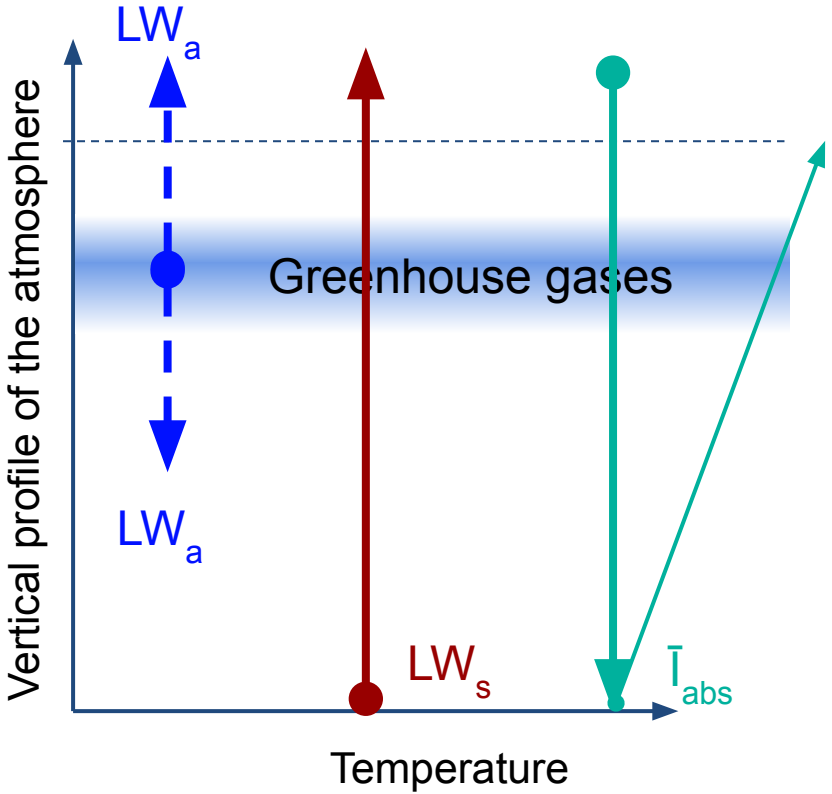
Adding one greenhouse layer

- Now we add greenhouse gases
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$$T_\mathrm{s} = \left(\frac{2\,\overline{I}_\mathrm{abs}}{\sigma}\right)^{1/4} = 2^{1/4}\left(\frac{\overline{I}_\mathrm{abs}}{\sigma}\right)^{1/4}$$

$$= 2^{1/4}\,T_\mathrm{e}$$

$$\approx 1{,}189 \cdot 255\,[\mathrm{K}] \approx 303\,[\mathrm{K}] \approx +30\,^\circ\mathrm{C}$$



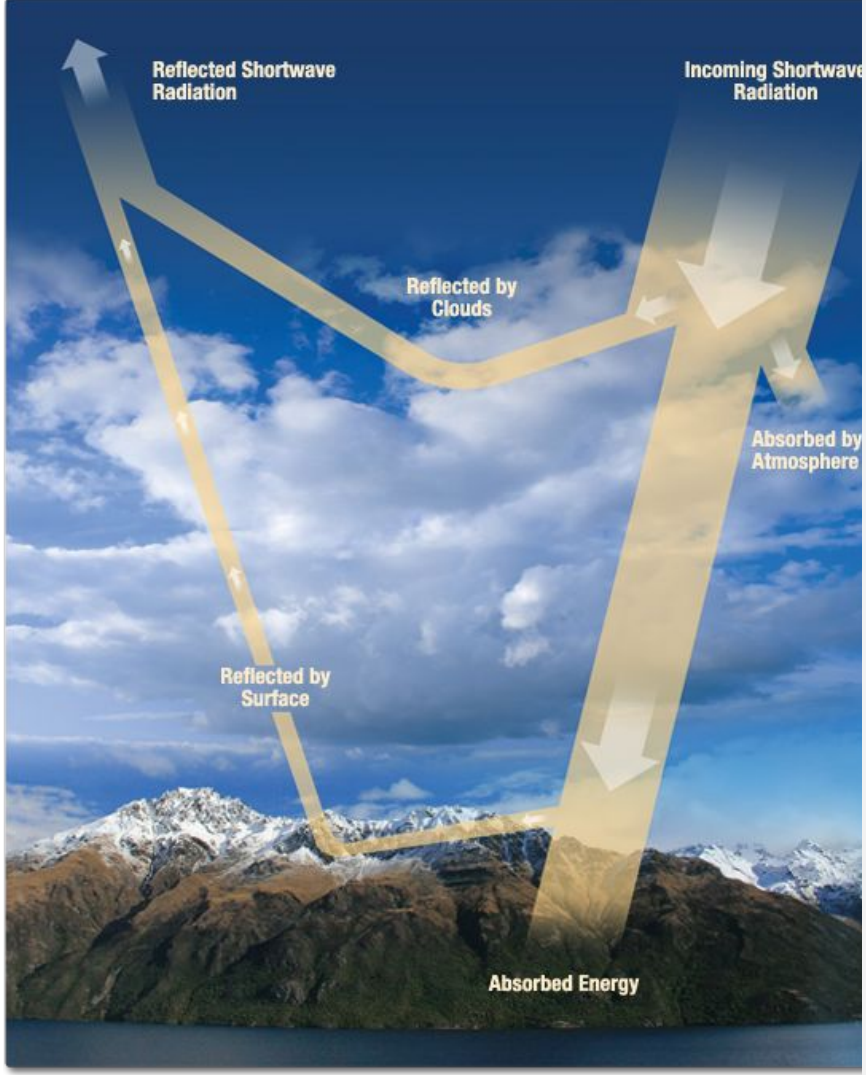
No atmosphere: −18 °C

Perfect absorber: +30 °C

Reality: +15 °C

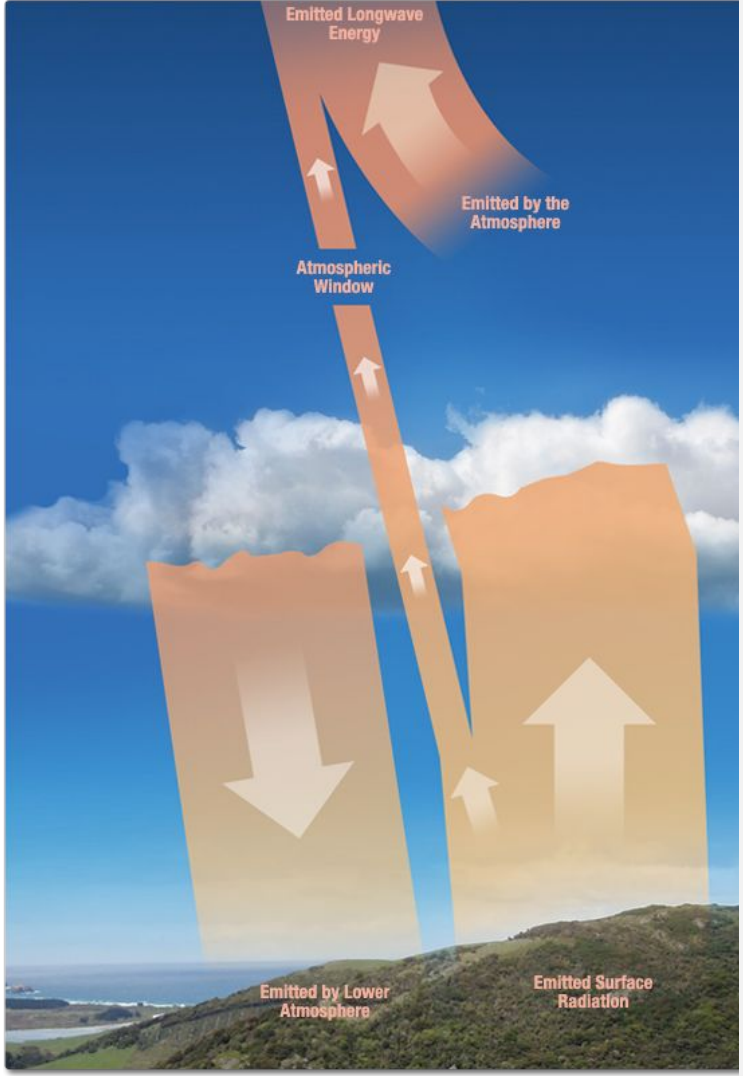
Which assumption is too strong?

What happens to the incoming shortwave radiation?



https://science.nasa.gov/ems/13_radiationbudget

What happens to the outgoing longwave radiation?



https://science.nasa.gov/ems/13_radiationbudget

Candidate 1: the atmosphere is not a perfect absorber ($\varepsilon_a <$

Formula sheet I-13

Atmosphere: $\varepsilon_a \sigma T_s^4 = 2\,\varepsilon_a \sigma T_a^4 \quad \Rightarrow \quad T_a^4 = \frac{1}{2} T_s^4 \qquad (\varepsilon_a \sigma \text{ cancels})$

Surface: $\overline{I}_{abs} + \varepsilon_a \sigma T_a^4 = \sigma T_s^4$

insert T_a^4 : $\overline{I}_{abs} + \varepsilon_a \sigma \frac{1}{2} T_s^4 = \sigma T_s^4$

$$\overline{I}_{abs} = \sigma T_s^4 - \tfrac{\varepsilon_a}{2} \sigma T_s^4 = \sigma T_s^4 \left(1 - \tfrac{\varepsilon_a}{2}\right)$$

$$T_s^4 = \frac{\overline{I}_{abs}}{\sigma} \cdot \frac{1}{1 - \varepsilon_a/2}$$

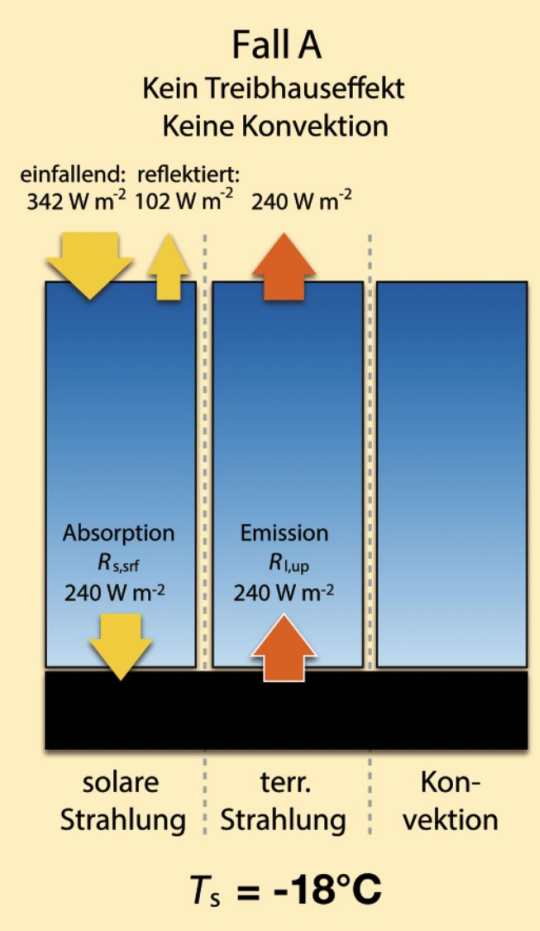
$$T_s = \underbrace{\left(\frac{\overline{I}_{abs}}{\sigma}\right)^{1/4}}_{T_e \approx 255\,[\mathrm{K}]} \cdot \left(1 - \frac{\varepsilon_a}{2}\right)^{-1/4} = T_e \left(1 - \frac{\varepsilon_a}{2}\right)^{-1/4}$$

ε_a	atmospheric emissivity/absorptivity in the longwave, $0 \le \varepsilon_a \le 1$ $[-]$
$\varepsilon_a = 1$	perfect blackbody atmosphere: $T_s = 2^{1/4} T_e \approx 303\,[\mathrm{K}]$ (too warm)
$\varepsilon_a \approx 0.78$	best fit to observed $T_s \approx 288\,[\mathrm{K}]$: $(1 - 0.39)^{-1/4} \approx 1.13$
$\varepsilon_a = 0$	no greenhouse effect: $T_s = T_e \approx 255\,[\mathrm{K}]$ (too cold)

The atmospheric window (8–12 µm) lets some surface LW escape directly to space, so $\varepsilon_a < 1$ in reality. Tuning ε_a to match $T_s = 288\,[\mathrm{K}]$ gives $\varepsilon_a \approx 0.78$. However, ε_a is a phenomenological fit – the remaining gap ($303 \to 288\,[\mathrm{K}]$) also reflects non-radiative surface cooling (latent heat, sensible heat), which we add next.

But there could be another reason!

Recall the energy balance



Kleidon, A. (2020). Sonne oder Treibhauseff
Globale Erwärmung einfach und physikalisch
nachgerechnet. *Physik in unserer Zeit*, *51*(2)
79-85.

Step 1 – radiation only: 255 K, too cold

Absorbed radiation = emission of the Earth (expected as longwave infrared radiation)

$$\underbrace{\sigma T_s^4}_{LW_\uparrow} = \underbrace{\frac{I_0}{4}(1-\alpha)}_{SW_\downarrow - SW_\uparrow}$$

Formula sheet I-11

$$T_s = \left(\frac{SW_\downarrow - SW_\uparrow}{\sigma}\right)^{1/4} = \left(\frac{\frac{1}{4}I_0(1-\alpha)}{\sigma}\right)^{1/4} \approx 255\,[\mathrm{K}] \approx -18\,^\circ\mathrm{C}$$

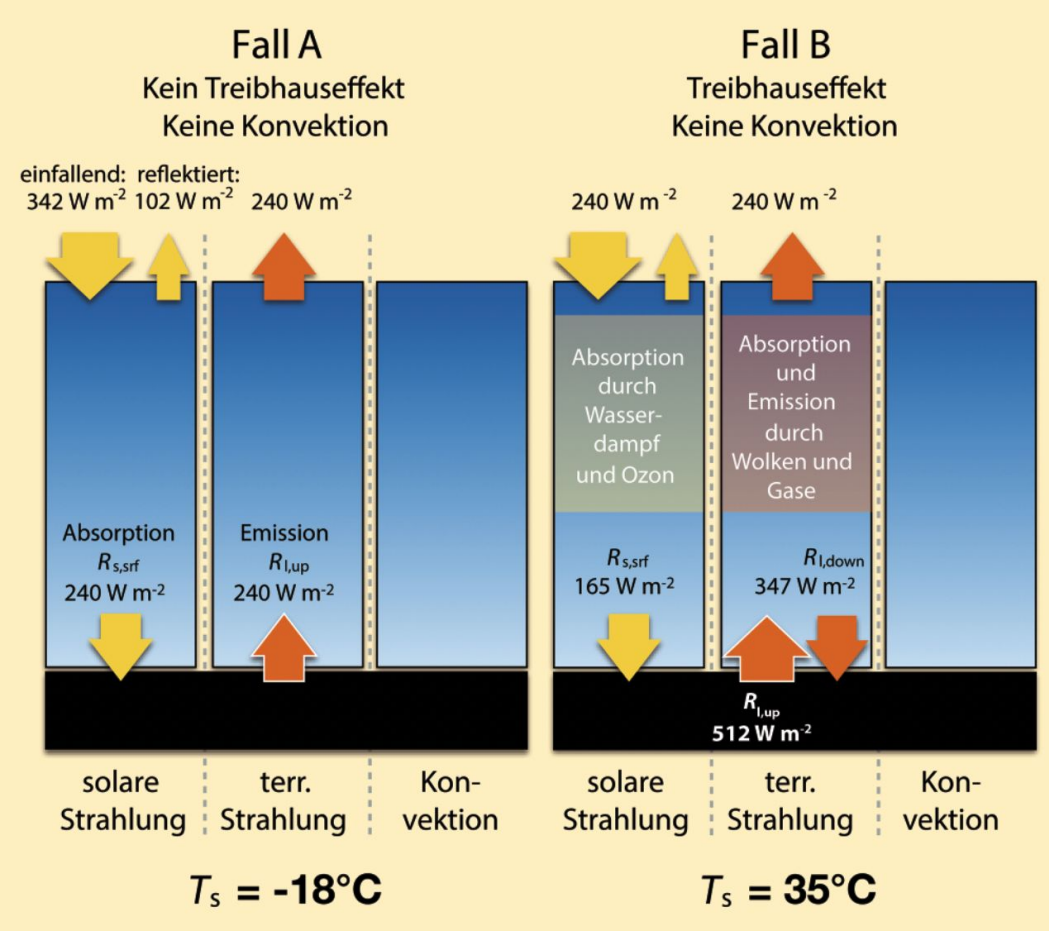
$LW_\uparrow = \sigma T_s^4$ upwelling longwave from surface $[\mathrm{W\,m^{-2}}]$

$SW_\downarrow - SW_\uparrow$ net absorbed shortwave $= \frac{I_0}{4}(1-\alpha) \approx 238\,[\mathrm{W\,m^{-2}}]$

$T_s \approx -18\,^\circ\mathrm{C}$ too cold: atmosphere and $LW_\downarrow$ missing

Baseline: surface radiates away everything it absorbs from the Sun. No atmosphere, no back-radiation, no turbulent fluxes. This is the T_e from I-11.

Adjusting the balance



Kleidon, A. (2020). Sonne oder Treibhauseff
Globale Erwärmung einfach und physikalisch
nachgerechnet. *Physik in unserer Zeit*, *51*(2)
79-85.

Step 2 – adding back radiation: 306 K, too warm

$$\underbrace{\sigma T_s^4}_{LW_\uparrow} = \underbrace{SW_\downarrow - SW_\uparrow}_{\approx 161\,[\mathrm{W\,m^{-2}}]} + \underbrace{LW_\downarrow}_{\approx 333\,[\mathrm{W\,m^{-2}}]}$$

$$T_s = \left(\frac{SW_\downarrow - SW_\uparrow + LW_\downarrow}{\sigma}\right)^{1/4} \approx \left(\frac{494\,[\mathrm{W\,m^{-2}}]}{5.670 \times 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}]}\right)^{1/4} \approx 306\,[\mathrm{K}] \approx +32\,^\circ\mathrm{C}$$

$SW_\downarrow - SW_\uparrow$ net shortwave absorbed *at the surface*, $\approx 161\,[\mathrm{W\,m^{-2}}]$ (the atmosphere itself absorbs ≈ 78 of the $239\,[\mathrm{W\,m^{-2}}]$)

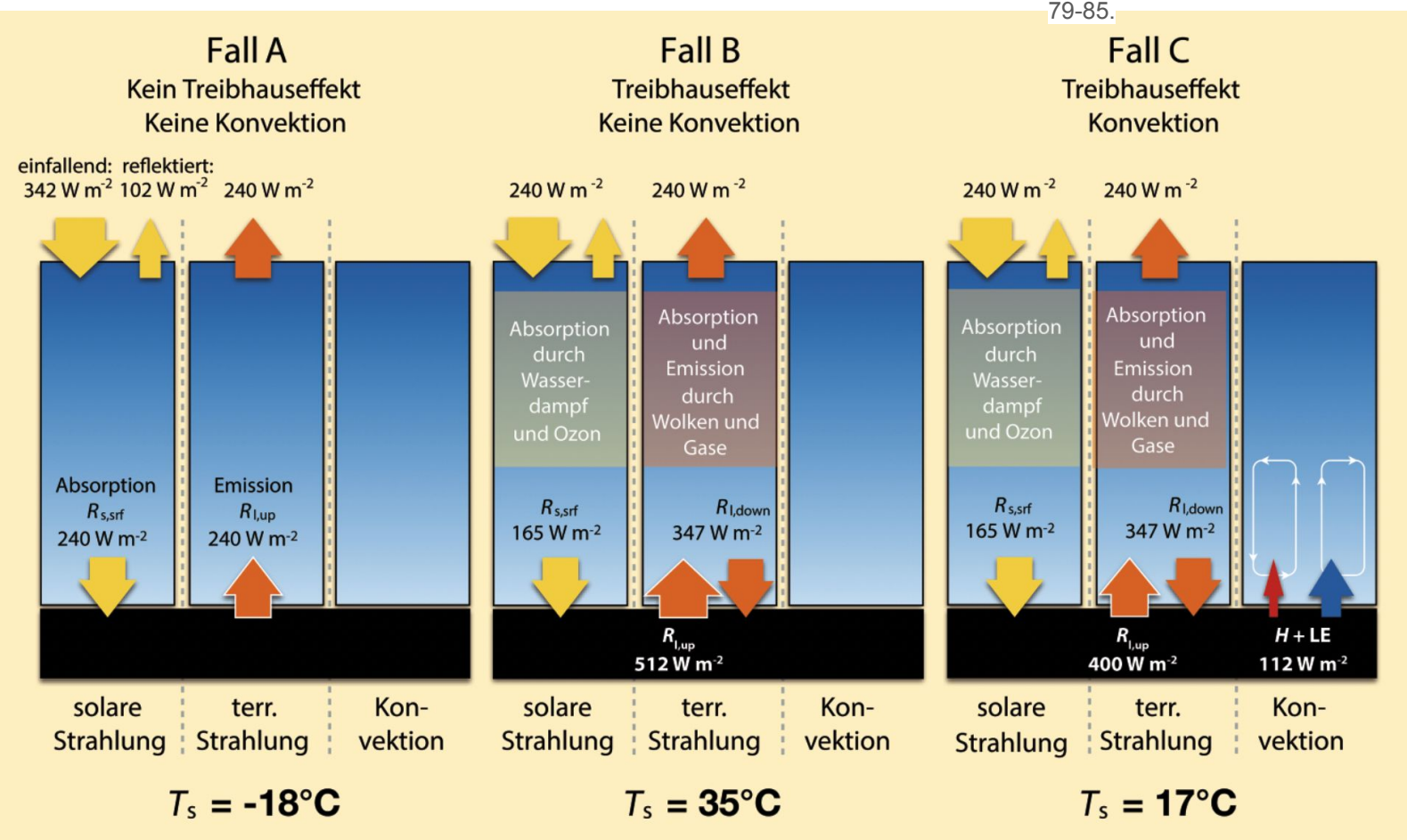
$LW_\downarrow$ downwelling longwave (atmospheric back-radiation), $\approx 333\,[\mathrm{W\,m^{-2}}]$

$T_s \approx +32\,^\circ\mathrm{C}$ now too warm: turbulent surface fluxes missing

Still wrong … something is missing here!

Radiation balance

Kleidon, A. (2020). Sonne oder Treibhauseff
Globale Erwärmung einfach und physikalisch
nachgerechnet. *Physik in unserer Zeit*, *51*(2)
79-85.



Step 3 – adding H and LE: 289 K ≈ observed

- We only arrive at reasonable global values if, in addition to the greenhouse effect, we also take into account the cooling effect of turbulent heat fluxes.
- Turbulent heat fluxes consist of the sensible and latent heat fluxes *H* and *LE.*

$$\underbrace{\sigma T_s^4}_{LW_\uparrow} = \underbrace{SW_\downarrow - SW_\uparrow}_{\approx 161} + \underbrace{LW_\downarrow}_{\approx 333} - \underbrace{(H + LE)}_{\approx 97\,[\mathrm{W\,m^{-2}}]}$$

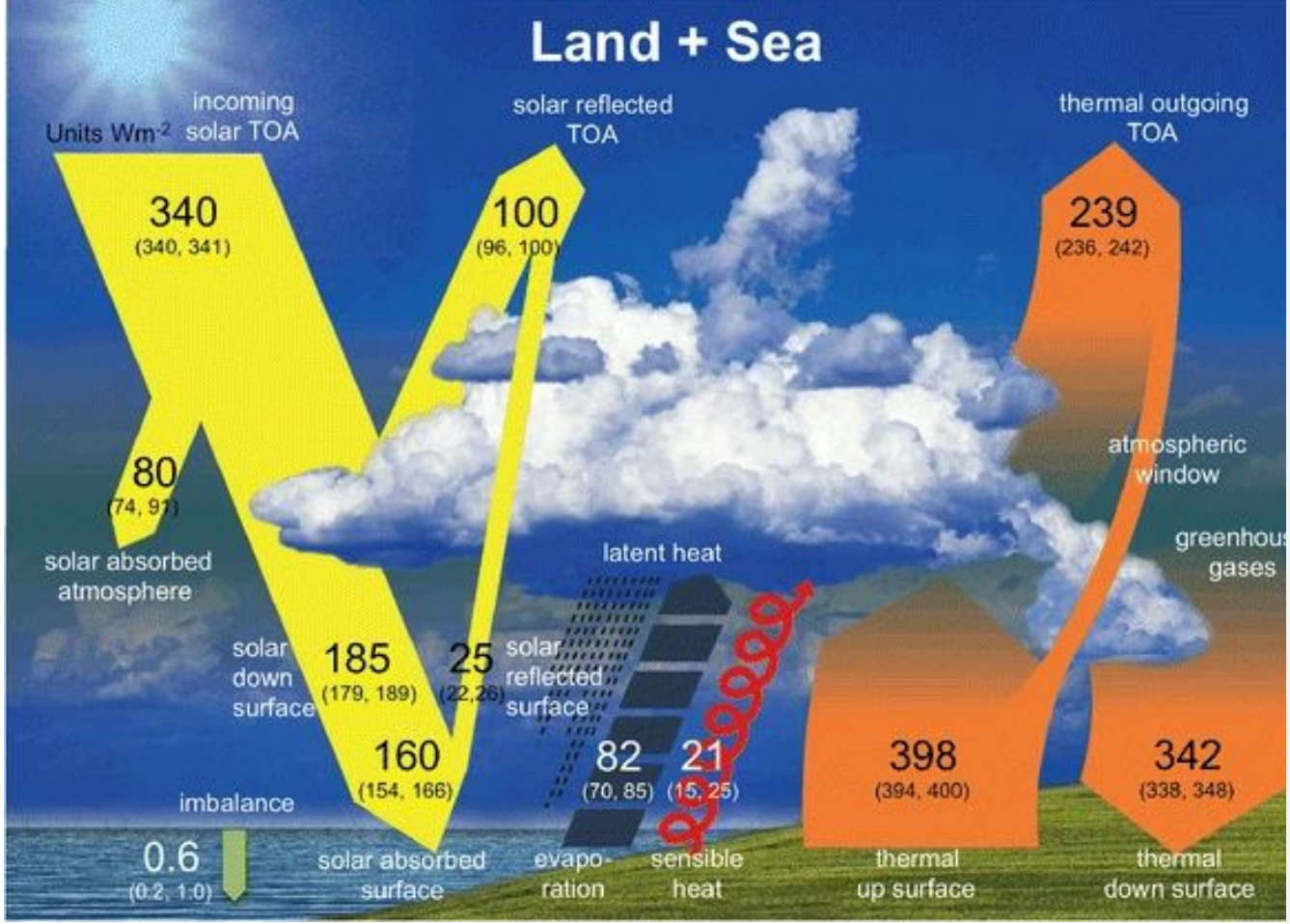
$$T_s = \left(\frac{SW_\downarrow - SW_\uparrow + LW_\downarrow - (H + LE)}{\sigma} \right)^{1/4}$$

$$\approx \left(\frac{397\,[\mathrm{W\,m^{-2}}]}{5.670 \times 10^{-8}\,[\mathrm{W\,m^{-2}\,K^{-4}}]} \right)^{1/4} \approx 289\,[\mathrm{K}] \approx +16\,^\circ\mathrm{C}$$

H	sensible heat flux (conduction + convection upward), $\approx 17\,[\mathrm{W\,m^{-2}}]$
LE	latent heat flux (evaporation), $\approx 80\,[\mathrm{W\,m^{-2}}]$
$H + LE$	total turbulent cooling, $\approx 97\,[\mathrm{W\,m^{-2}}]$ ($\approx 60\,\%$ of the net SW at the surface)

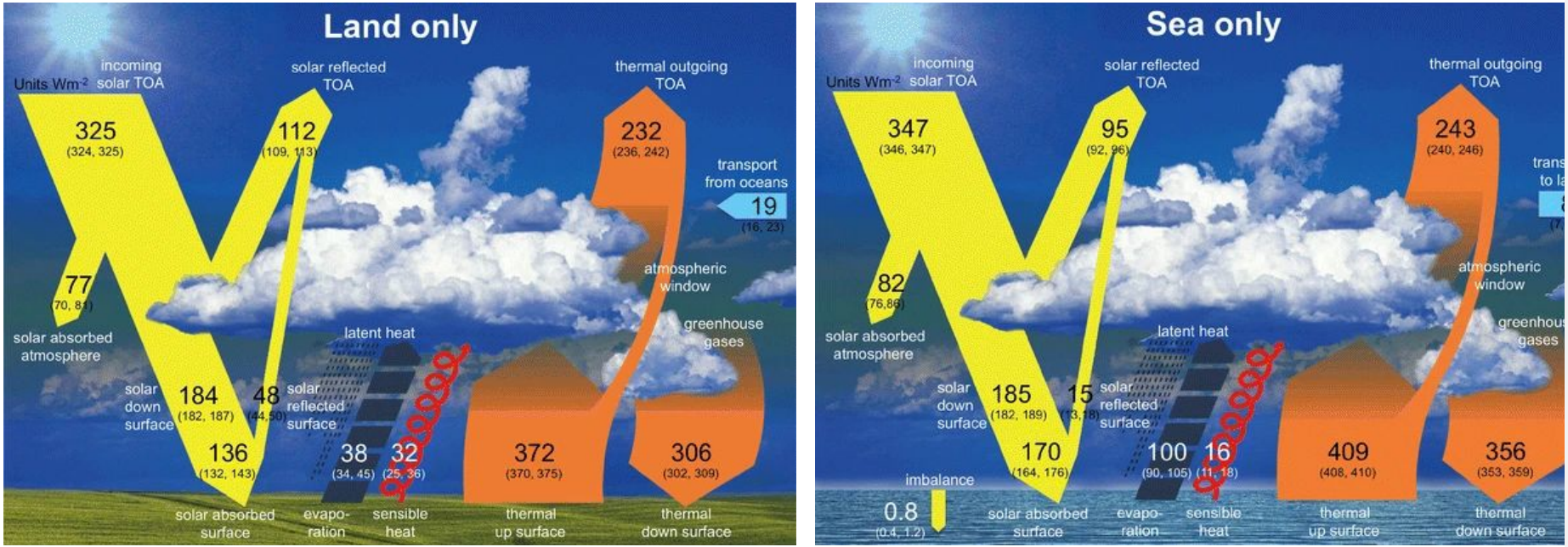
$T_s \approx +16\,^\circ\mathrm{C}$ close to observed $\approx +15\,^\circ\mathrm{C}$ ($LW_\uparrow \approx 396\,[\mathrm{W\,m^{-2}}]$)

Full balance



Wild, M., Folini, D., Hakuba, M.Z. *et al.* The energy balance over land and oceans: an assessment based on direct observations and CMIP5 climate models. *Clim Dyn* 44, 3393–3429 (2015)

Surface-dependent balance



Wild, M., Folini, D., Hakuba, M.Z. *et al.* The energy balance over land and oceans: an assessment based on direct observations and CMIP5 climate models. *Clim Dyn* 44, 3393–3429 (2015)

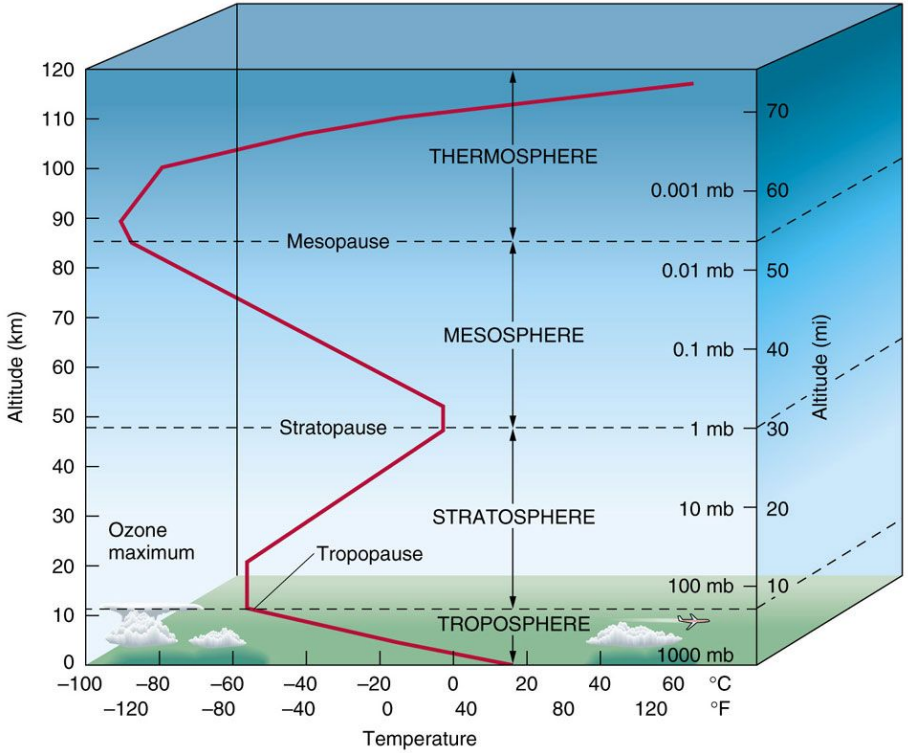
A full energy balance

Component	This study		
	Land[a]	Ocean[a]	Globe[a,b]
TOA			
Solar down	324.7	346.6	340.4
Solar up	−111.4	−94.9	−99.7
Solar net	213.3	251.7	240.7
Thermal up	−232.4	−242.9	−239.8
Atmosphere			
Solar net	77.3	81.7	80.7
Thermal net	−166.4	−189.9	−183.8
Surface			
Solar down	184	185	184.7
Solar up	−48	−15	−24.6
Solar net	136	170	160.1
Thermal down	306	356	341.5
Thermal up	−372	−409	−398.2
Thermal net	−66	−53	−56.7
Net radiation	70	117	103.3
Latent heat	−38	−100	−82.0
Sensible heat	−32	−16	−20.7

Wild, M., Folini, D., Hakuba, M.Z. *et al.* The energy balance over land and oceans: an assessment based on direct observations and CMIP5 climate models. *Clim Dyn* 44, 3393–3429 (2015)

STRUCTURE OF THE ATMOSPHERE

Vertical structure of the atmosphere

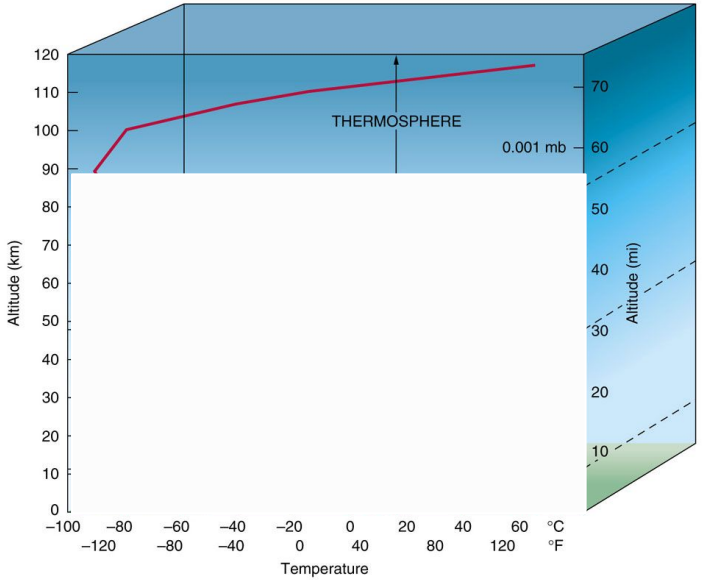


- The vertical temperature profile is far more complex than that of pressure.
- The atmosphere is “layered”, although we speak of a gas envelope.
- Each “floor” has its own characteristics, particular processes, and relevance for the functioning of the Earth system

THERMOSPHERE

Thermosphere

- Molecules are dissociated and ionised by X-ray and UV radiation → gases in the thermosphere exist predominantly as a plasma of ions, electrons and neutral particles.
- (+)Height → (+)Radiation intensity → (-)Recombination rate → (+)Ionisation degree.
- Light particles are faster and therefore less influenced by gravity → accumulation of light atoms and ions in the upper part of the thermosphere.



Just to give you an idea of the proportions …



Own images; Tromso 2026

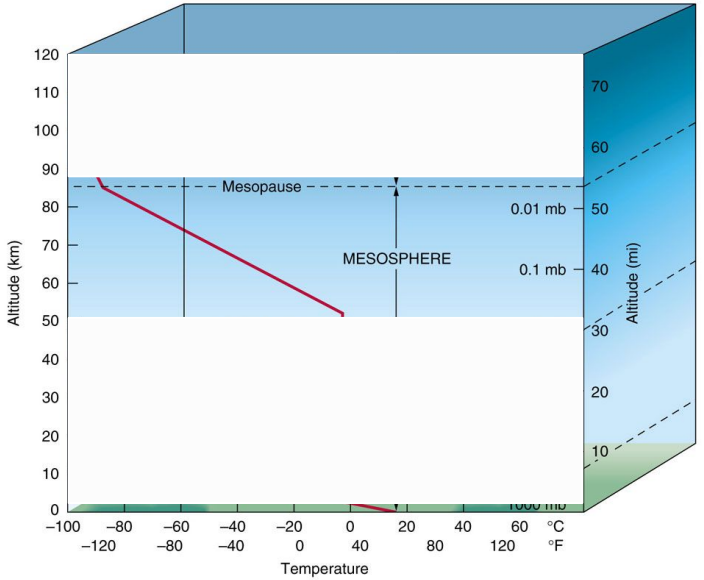


MESOSPHERE

Mesosphere

- Temperature drops from about 0 °C at the stratopause to an average of about −90 °C above 80 km.

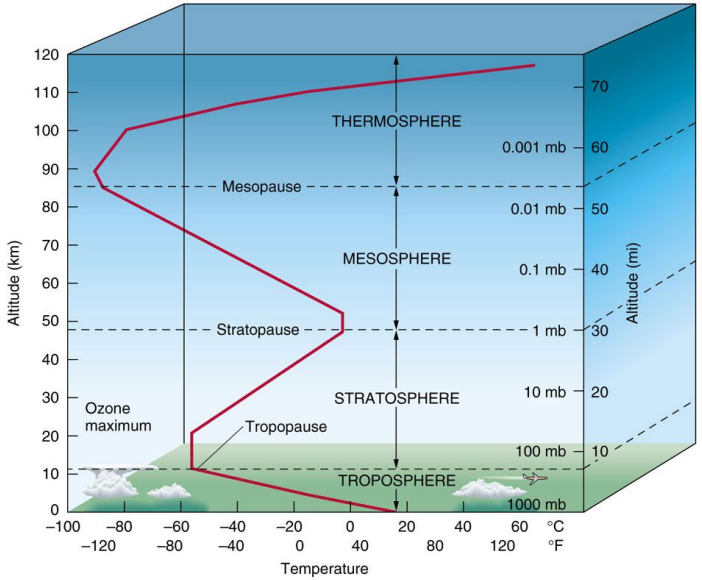
- Since the mesosphere lies above the ozone layer, the UV light here is still so strong that a human body would suffer severe burns within a very short time.



Stratopause

- Boundary between stratosphere and mesosphere.

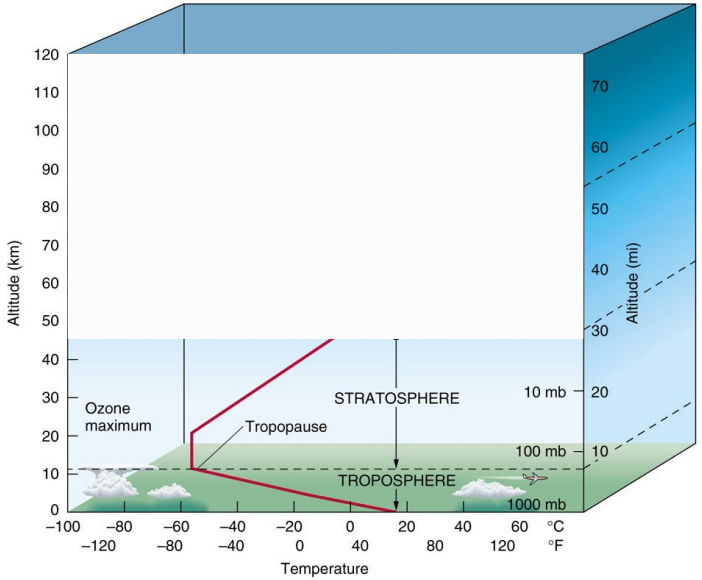
- Characterised by a pronounced maximum of the atmospheric temperature profile at an altitude of about 50 km.



STRATOSPHERE

Stratosphere: blanket layer

- O3 is formed here by photochemical processes, with a natural maximum at approx. 20–30 km.
- Important function: Absorption of UV radiation (up to 97 %).
- Radiation absorption → consequently, temperature increases with height
- Due to the temperature stratification, vertical exchange is strongly suppressed (more on this in the next lecture).



Formation and destruction of ozone – the ozone cycle

High-energy, very short-wave radiation (UV-C) splits O_2.

$$O_2 \xrightarrow{h \cdot v} 2\,O \quad (\lambda \leq 242\,\mathrm{nm})$$

Oxygen radicals react to form ozone. Collision partner “M” carries away the energy → heating of the environment.

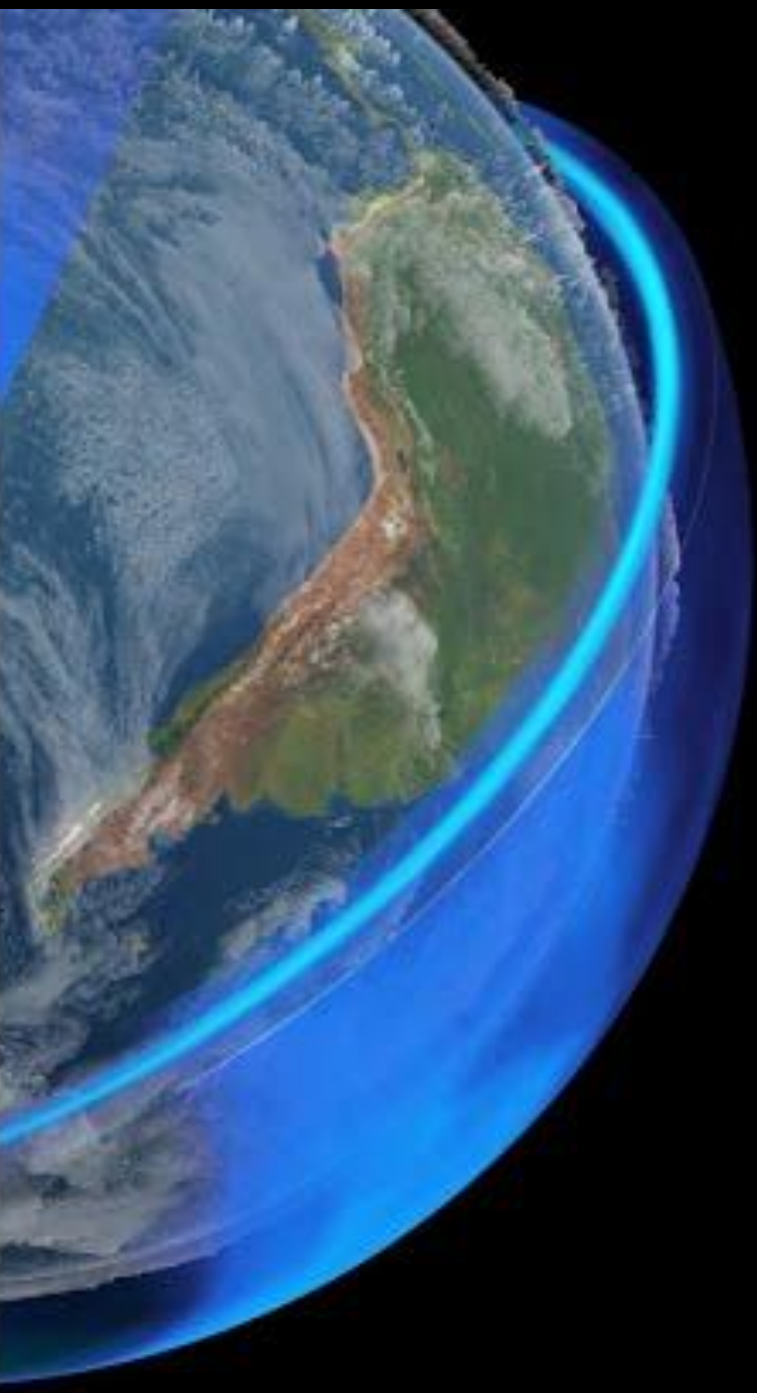
$$O_2 + O + M \longrightarrow O_3 + M$$

Ozone formation is always accompanied by ozone destruction, which is also driven by UV radiation.

$$O_3 \xrightarrow{h \cdot v} O + 2\,O \quad (\lambda \leq 310\,\mathrm{nm})$$

Together with water, hydroxyl radicals are formed, which remove e.g. CO from the stratosphere.

$$H_2O + O \longrightarrow 2\,OH$$



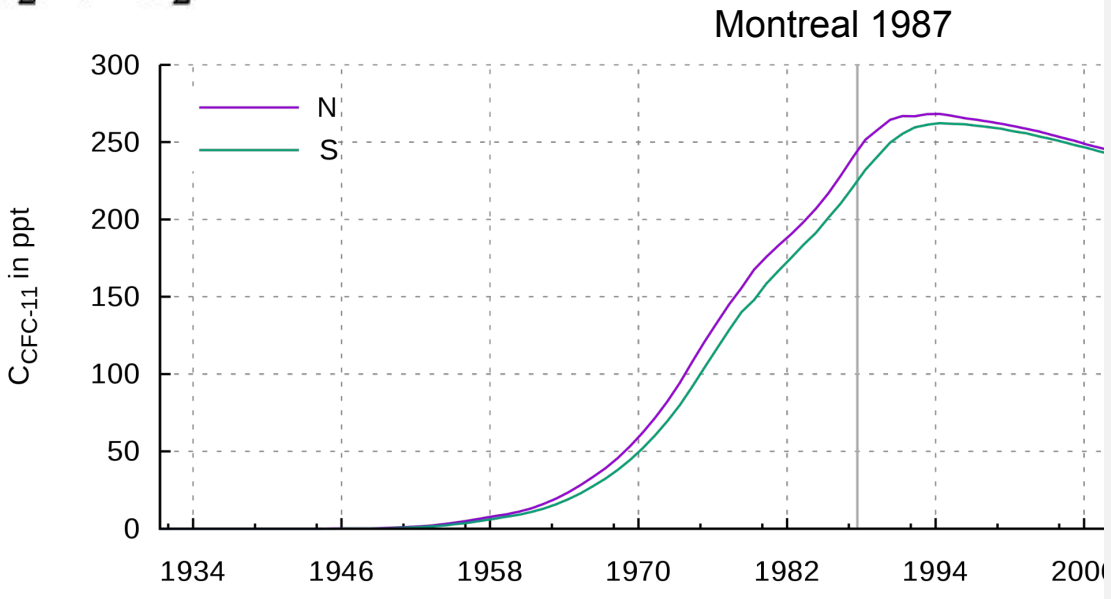
CFCs in the ozone layer

$$CF_2Cl_2 + h \cdot \nu \longrightarrow CF_2Cl^* + Cl^*$$

$$Cl^* + O_3 \longrightarrow ClO^* + O_2$$

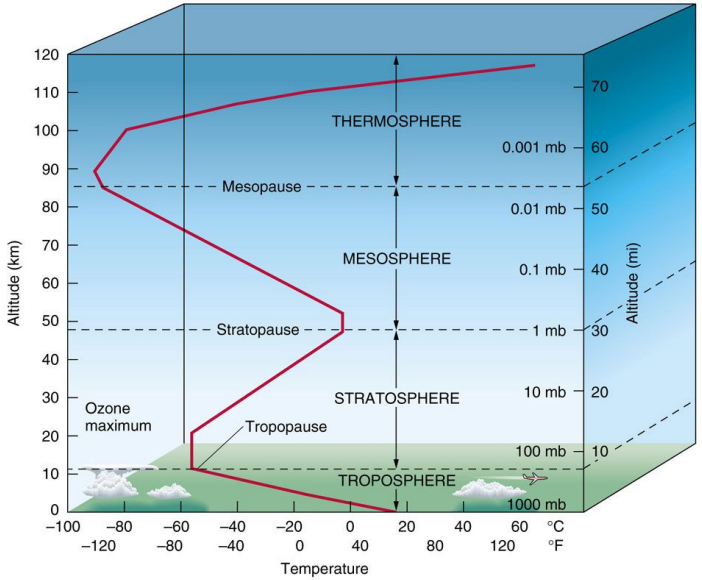
$$2\,ClO^* \longrightarrow Cl_2O_2 \longrightarrow Cl_2 + O_2$$

$$Cl_2 + h \cdot \nu \longrightarrow 2\,Cl^*$$



Tropopause: boundary between the turbulent troposphere and the stable stratosphere

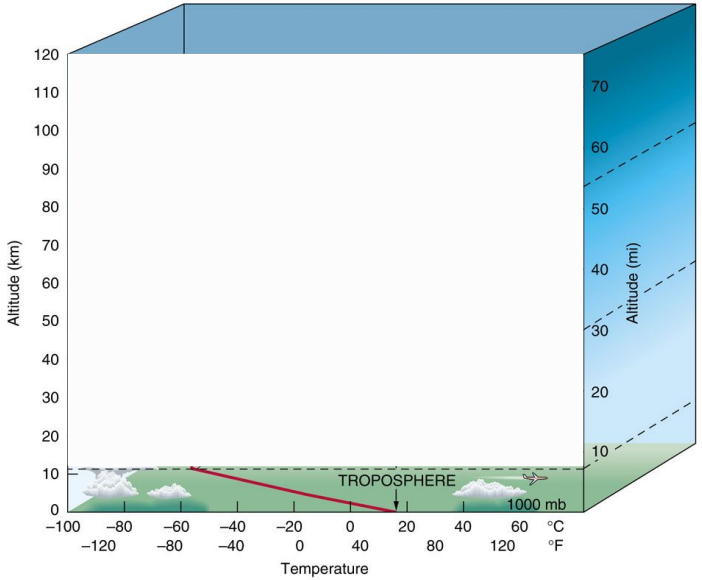
- Discovered (1902) because temperature suddenly stops decreasing with height
- Above it: ozone increases, water vapour drops sharply



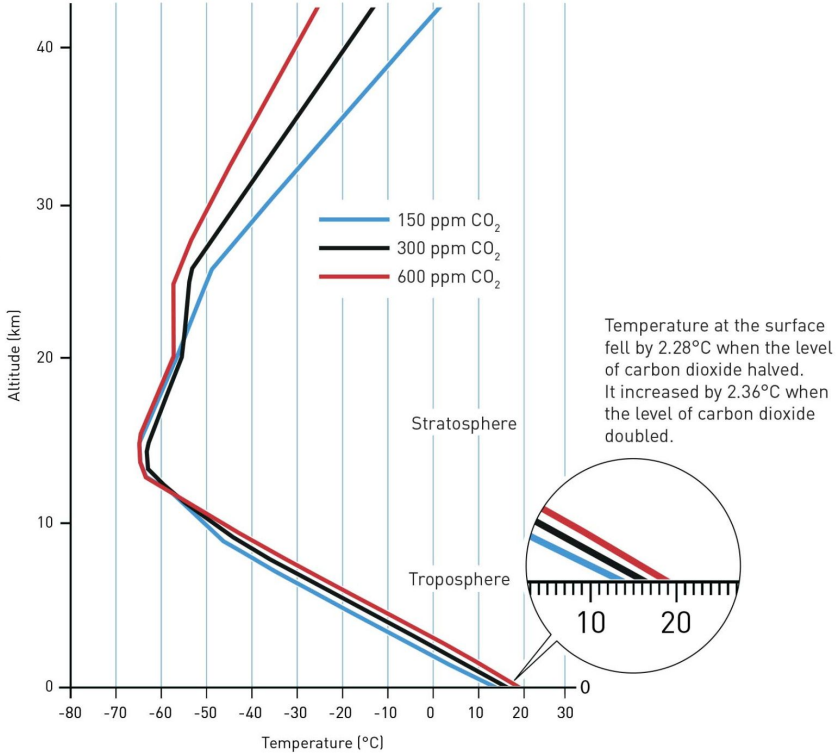
TROPOSPHERE

Troposphere: “motion”

- Where “weather happens”
- Only a small fraction of the volume …
- … but about 80 % of the mass of the atmosphere
- Comprises the lowest 8 to 16 km (poles to tropics)
- Temperature decreases with altitude
- At the bottom: the “planetary boundary layer”, where exchange with the land surface and oceans takes place



Can you explain this pattern? What does it mean?



Source: Manabe and Wetherald (1967) Thermal equilibrium of the atmosphere with a given distribution of relative humidity, *Journal of the atmospheric sciences*, Vol. 24, Nr 3, May.

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PORTFOLIO QUESTION

Q2: Please answer one of the following questions

1. Why are certain gases called “greenhouse gases” and what are the exact physical mechanisms behind that?

2. How does ozone interact with UV radiation i.e. what is the physical-chemical reaction behind that interaction?

3. What is the exact explanation for the emergence of the aurora borealis and what colours are caused how?